



Digital AC-DC Electropenetrograph (EPG)

Auburn University & USDA

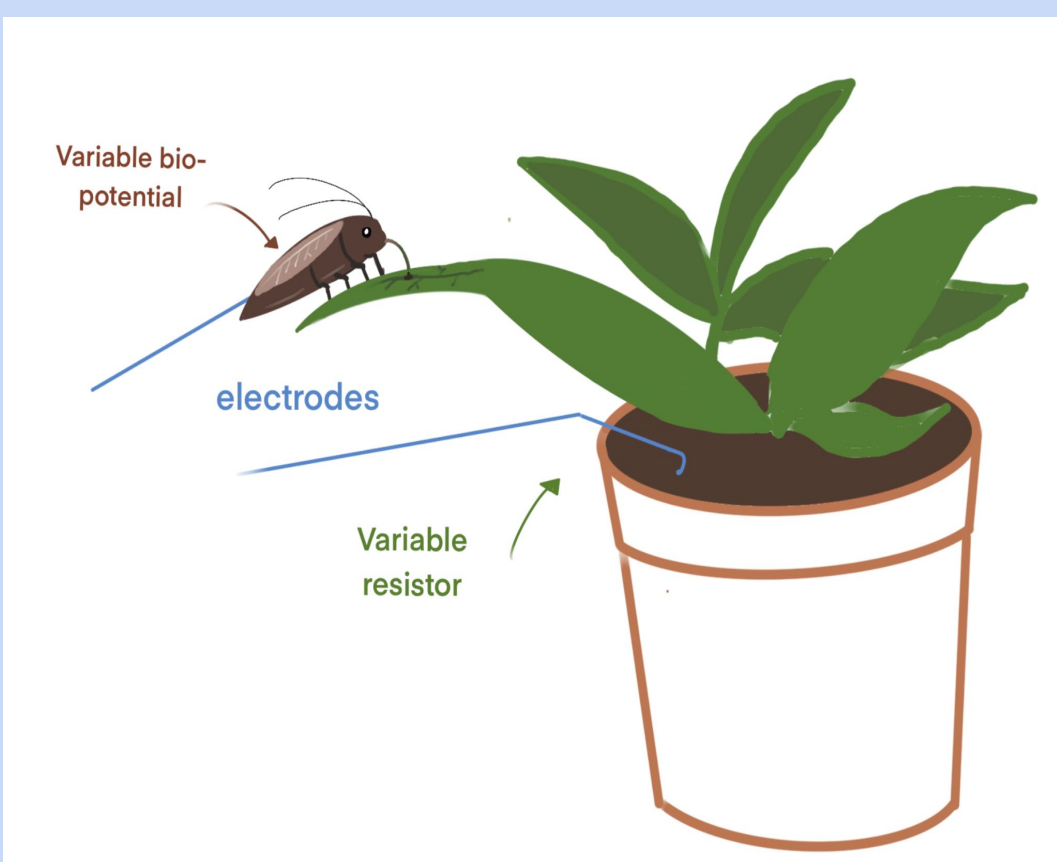
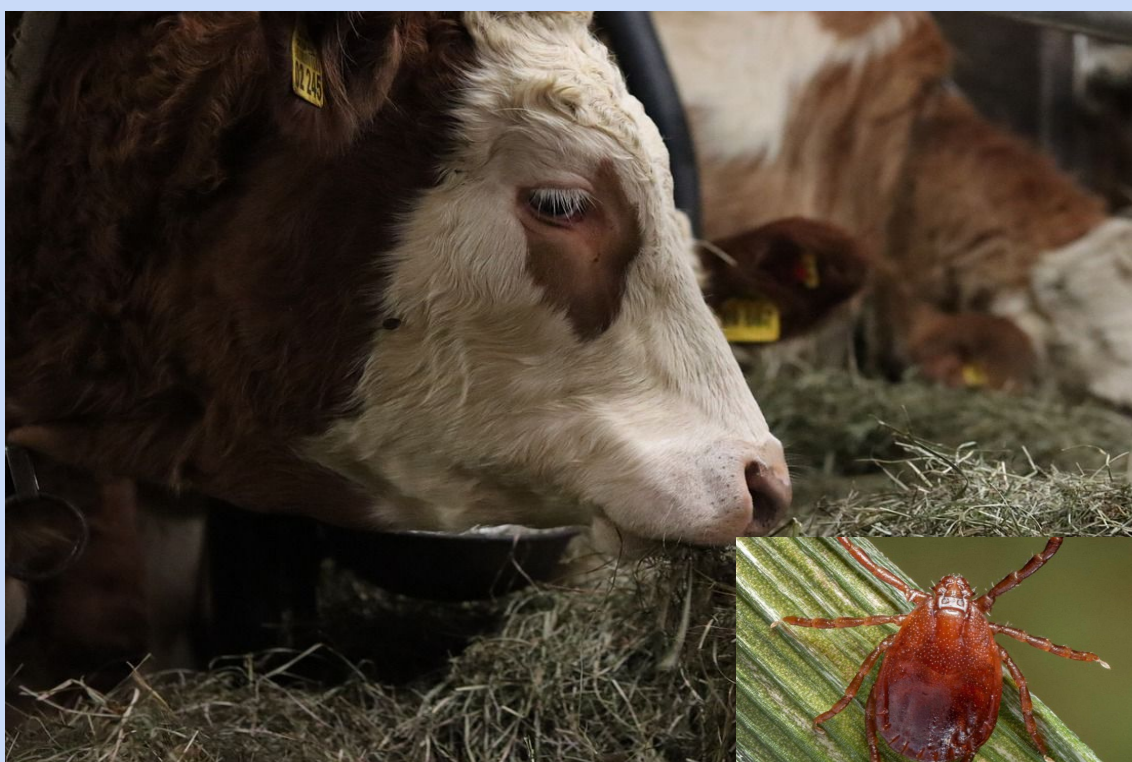
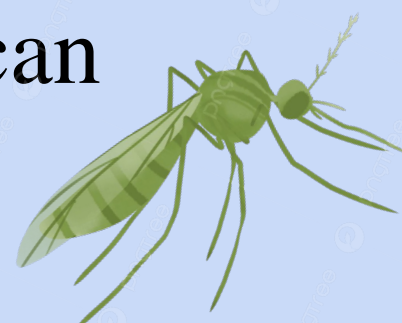


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Faculty Advisor: Prof. Matthew Spencer (F) and Prof. Victor Shia (S)



Arthropod-borne diseases harm crops, livestock, and people

- Asian Citrus Psyllid spreads citrus greening and threatens \$3.3 Billion worth of citrus production in the US [1]
- More than 700,000 deaths from vector-borne diseases are caused annually by mosquitoes [2]
- EPG allows researchers to study the feeding behavior of these arthropods so that preventive measures can be taken, such as effectively designing pesticides.



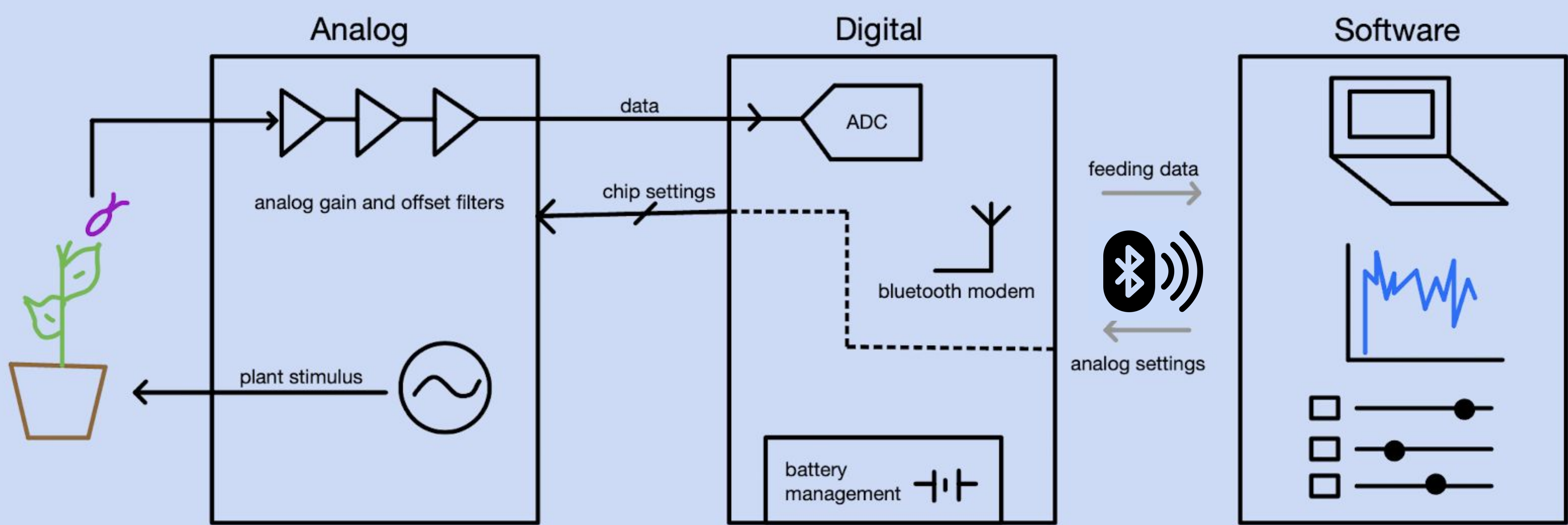
Physical bug-plant system set-up

EPG allows study of arthropod-borne disease transmission [3]

Working principle: Arthropod becomes an electrical component in a circuit

1. Apply excitation voltage arthropod
2. Arthropod electrical properties fluctuate during feeding
3. Measured voltage output waveform contains information that researchers use about feeding behavior

System Overview



Difficulties with Current EPG

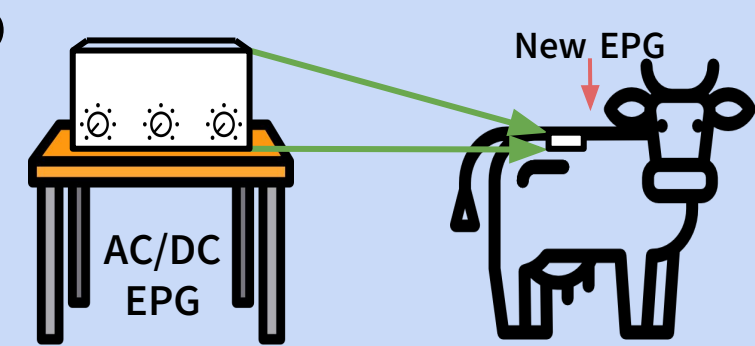
- Large livestock animals are required for most blood-feeding arthropod AC-DC EPG models
- Bulky size and wired wall power makes experimentation with large livestock animal research difficult.



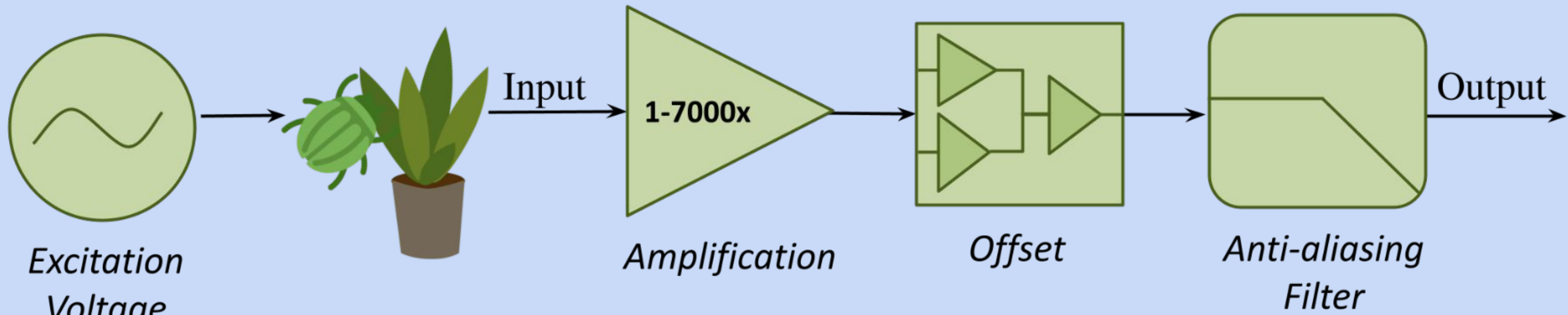
EPG control box for gain/offset settings and Faraday cage

Clinic Goals

1. Shrink AC-DC EPG to ensure field compatibility
2. Integrate wireless communication
3. Increase signal to noise ratio
4. Enhance user experience
5. Battery power

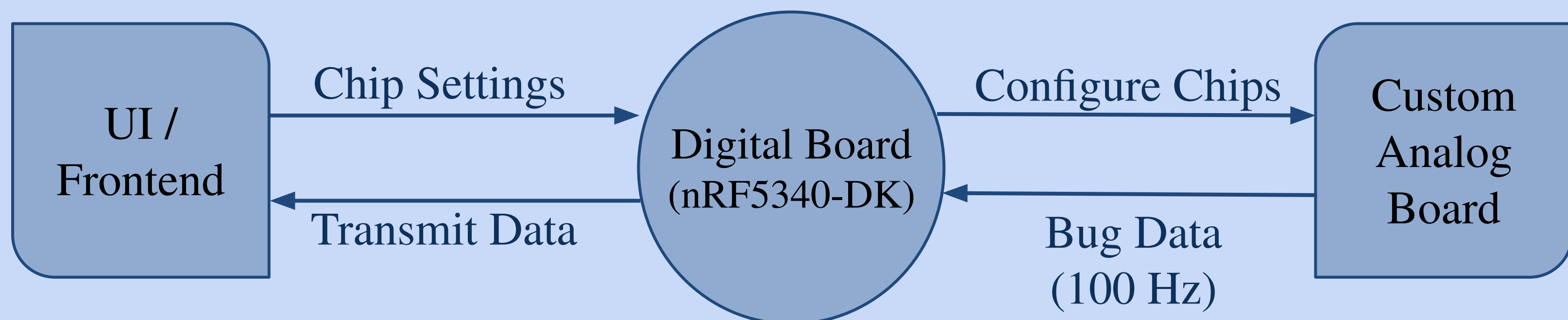


Analog Architecture



1. Excitation Voltage of 1000 Hz (AC) or 0 Hz (DC)
2. Resistance selection (up to 10 Gohms) depending on the bug under test
3. Amplify the bug waveform by up to 7000x
4. Offset the bug waveform to a readable range of the ADC
5. 1st order low pass filter to minimize aliasing

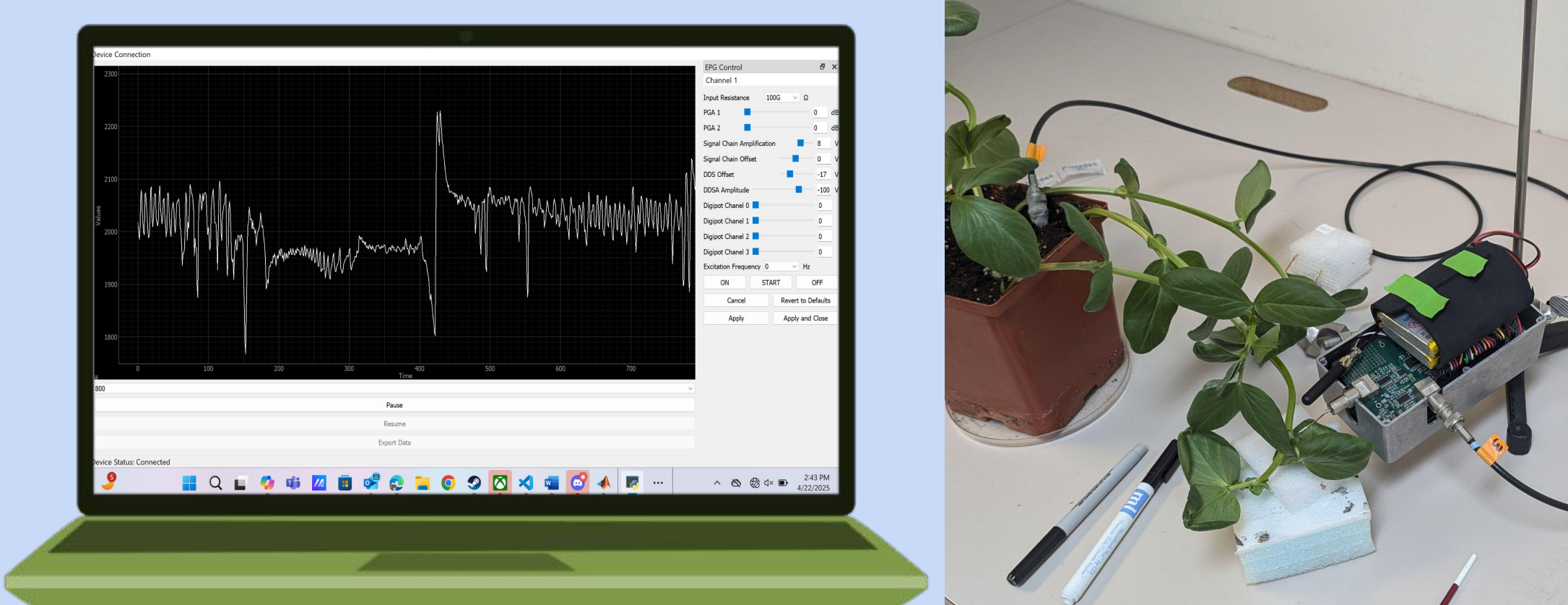
Software Architecture



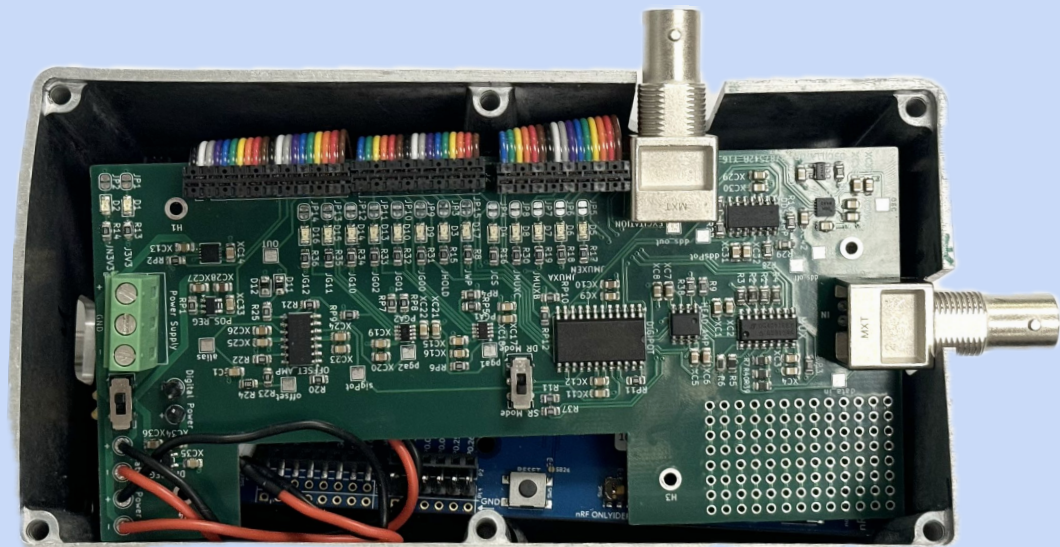
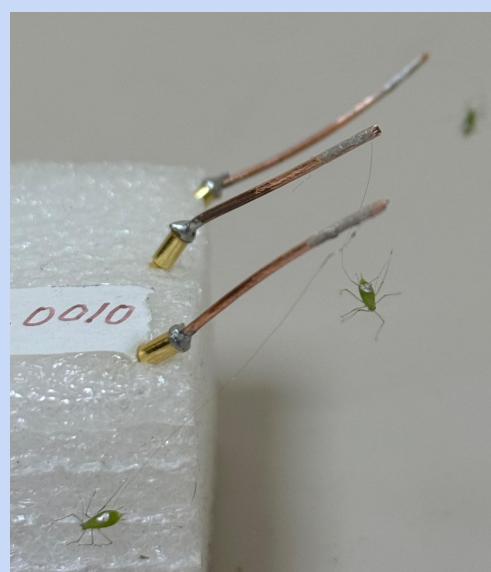
The digital system is controlled by two software components for real-time operation:

- **Frontend:** Provides a user interface to control chip settings and visualize data in real-time via Bluetooth.
- **Firmware:** Runs on the digital board to configure hardware components on the analog board such as excitation voltage, gain, and offset. It also receives and transmits data from the analog board to the UI via Bluetooth.

Experimental Setup

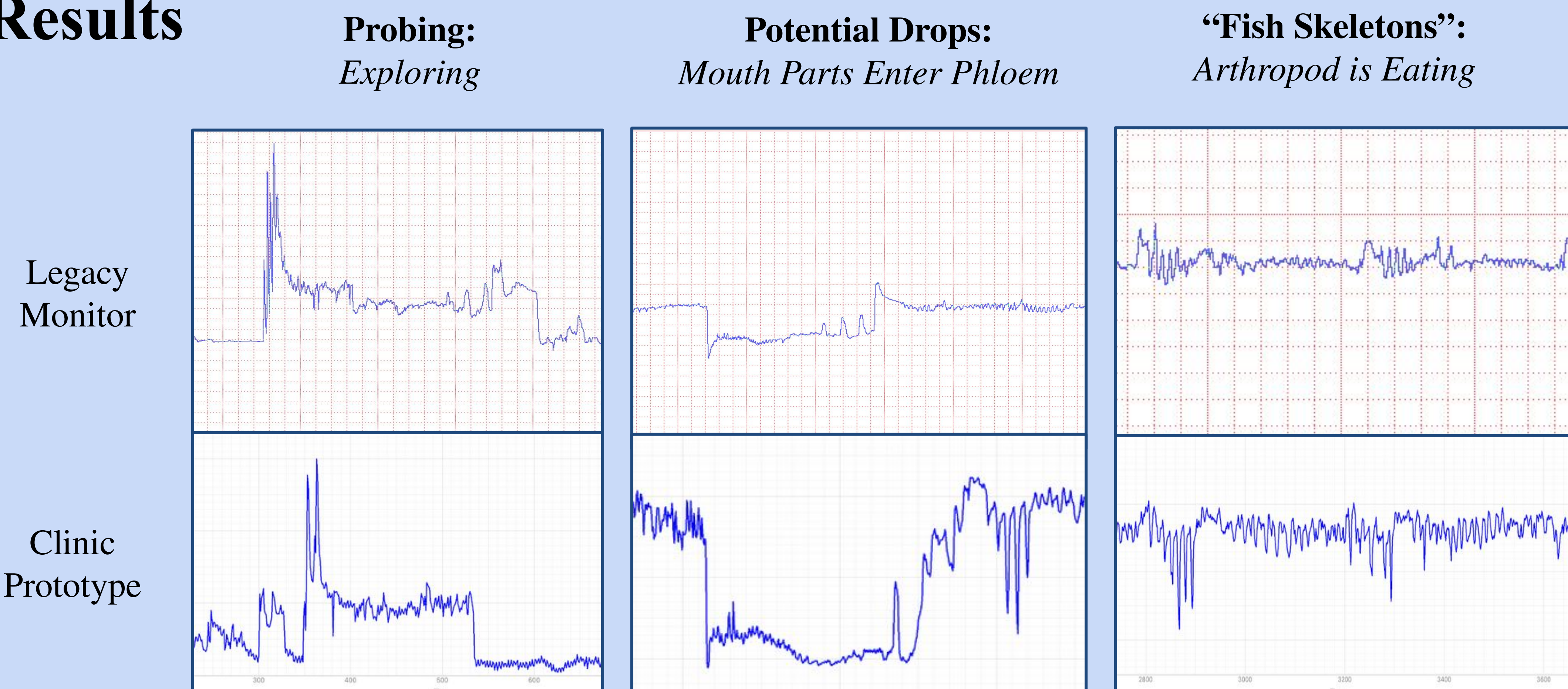


Left: 2025 UI, Right: Experimental setup with aphid, plant, and 2025 AC-DC EPG



Left: wired aphid, Right: closeup of 2025 AC-DC EPG

Results



Conclusion

1. Successful site visits to USDA research center in Parlier, CA showed comparable traces of the clinic prototype to the legacy monitor for pea aphids.
2. Reliable Bluetooth transmission range up to 46 feet through lab walls.
3. Accessible User Interface for entomologists to interface with the prototype.

Acknowledgements and References

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[1] "FINDINGS REGARDING AN EMERGENCY PROGRAM FOR ASIAN CITRUS PSYLLID AND HUANGLONGBING." Program CE-2943 , Orange County, Aug. 23, 2024
[2] A. Richter-Boix, "149 billion: the economic cost of invasive mosquitoes," Mosquito Alert. Accessed: Sep. 01, 2024. [Online]. Available: <https://www.mosquitoalert.com/en/149-000-millones-el-coste-economico-de-los-mosquitos-invasores/>
[3] Rafael Burger (TL-S), August McIntosh-Lombardo (F), Tjaard Van Loben Sels, Audrey Vo (S), and Simon Woodside (TL-F), Auburn-USDA Engineering Clinic Project Final Report 23-24, Unpublished document, Engineering, Harvey Mudd College, Claremont, California, 2024.